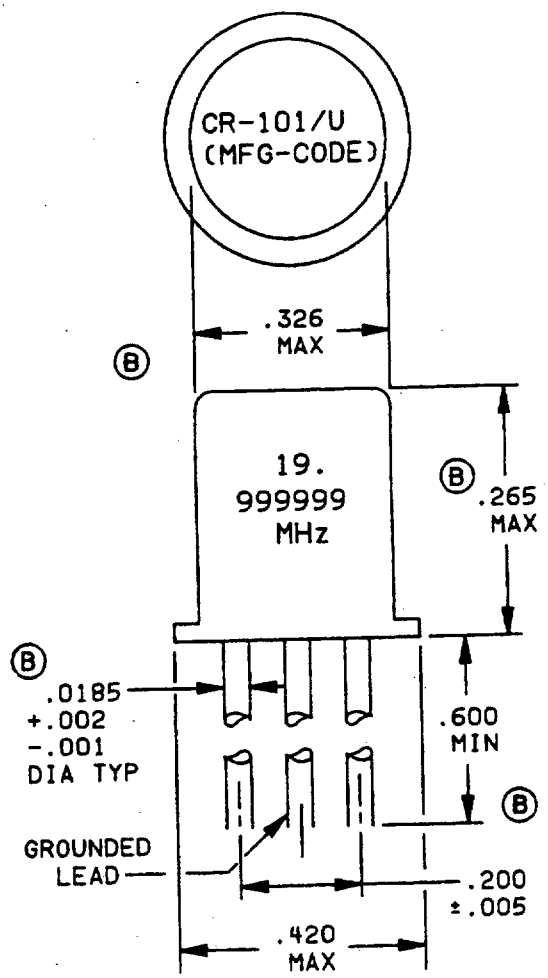
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-101/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 7 to 20 MHz; fundamental; noncontrolled; antiresonance; cold weld sealed



Inches	mm
.001	0.03
.002	0.05
.005	0.13
.0185	0.47
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Crystal unit - CR-101/U.

(B) denotes changes

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-35/U (MIL-H-10056/21).
Dimensions, marking, and configuration	See figure 1
Frequency range	7 to 20 MHz, inclusive
Frequency tolerances	±0.005 percent
Equivalent resistance	7 to 10 MHz - 30 ohms. 10+ to 20 MHz - 25 ohms.
Antiresonance, load capacitance	30 ±0.5 pF
Mode of oscillation	Fundamental
Operating temperature range	-55°C to +105°C, inclusive
Test set, calibration values and rated drive level	See table II
Capacitance, shunt	7 pF, maximum.
Shock:	
Permitted frequency change	±5 ppm
Permitted equivalent resistance change	±10 percent
Thermal shock:	
Permitted frequency change	±5 ppm
Permitted equivalent resistance change	±10 percent
Vibration: 1/	
Permitted frequency change	±5 ppm
Permitted equivalent resistance change	±10 percent
Aging	25 ppm

1/ Method 204 of MIL-STD-202, test condition B.

Ⓑ TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volts	mW
TS-330/TSM	7 to 10	13	20	---	5 ±1.0
TS-683/TSM	10+ to 20	10	--	0.22	5 ±1.0

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Ⓑ Qualification inspection reinstated as of the date of this document.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activi
Army - ER

Agent:
DLA - ES

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